

ClaimInsight360 - Insurance Claims Analytics & Intelligence Platform

1. Introduction

ClaimInsight360 is a web-based **Claims Analytics & Intelligence Platform** designed for insurance carriers to analyze claims performance, detect trends, calculate KPIs, predict fraud/denials, monitor adjuster productivity, perform root-cause analysis, and provide real-time operational visibility into claims portfolios. It aggregates claims, policy, customer, financial, and external data to generate insights supporting claims managers, analysts, actuaries, and executives.

The platform uses a **REST API-based microservice architecture**, with **Angular/React** frontend and either **Java (Spring Boot)** or **.NET Core** backend.

Actors / Users

- **Claims Analyst:** Analyzes claim trends, denials, cost drivers.
- **Claims Manager:** Oversees performance metrics, delays, workloads, cycle time.
- **Fraud Analyst:** Uses risk dashboards and red-flag indicators.
- **Actuary:** Uses severity/frequency insights for pricing/reserving.
- **Operations Executive:** Monitors KPIs, TAT, compliance SLAs.
- **Admin:** Configures metrics, dashboards, data refresh, and roles.

2. Module Overview

2.1 Identity & Access Management

User authentication, RBAC, audit logs.

2.2 Data Ingestion & Claim Data Aggregator

Collects data from claims systems, policy systems, and external sources.

2.3 Claims Metrics Engine

Calculates KPIs such as TAT, severity, frequency, loss ratios, settlement times.

2.4 Fraud Indicators & Risk Scoring Engine

Risk signals, anomaly detection (rule-based), red-flag triggers.

2.5 Denial & Leakage Analysis

Analyzes denials, error codes, root-cause patterns, possible leakages.

2.6 Adjuster & Operations Performance Analytics

Evaluates workloads, productivity, cycle time, SLA violations.

2.7 Cost, Reserve & Trend Analytics

Tracks severity, reserves, aged claims, cost categories, inflation trends.

2.8 Claims Dashboard, Reports & Visualization

Interactive dashboards and drill-downs.

2.9 Notifications & Alerts

Insights alerts for anomalies, spikes, threshold breaches.

3. Architecture Overview

- **Frontend:** Angular or React
- **Backend:** REST API-based microservices
- **Data Store:** Relational DB + Analytical Warehouse (optional)
- **Integration:** Batch/API ingestion from Claims System

4. Module-Wise Design

4.1 Identity & Access Management Module

Features:

- Authentication, RBAC, audit logs

Entities:

- **User**
 - UserID
 - Name
 - Role (Analyst/Manager/Fraud/Actuary/Admin)
 - Email
 - Phone
- **AuditLog**
 - AuditID
 - UserID
 - Action
 - Resource
 - Timestamp
 - Metadata

4.2 Data Ingestion & Claim Data Aggregator Module

Features:

- Import claim, policy, payment, reserve, denial, fraud, adjuster data

Entities:

- **DataFeed**
 - FeedID
 - FeedType (Claim/Policy/Payment/Reserve)
 - SourceSystem
 - LastSyncDate
 - Status
- **ClaimRaw**
 - RawID
 - ClaimID
 - PayloadJSON
 - IngestedDate

4.3 Claims Metrics Engine Module

Features:

- Compute KPIs: TAT, cycle time, severity, frequency, loss ratio

Entities:

- **ClaimKPI**
 - KPIID
 - ClaimID
 - MetricName
 - MetricValue
 - MetricDate

4.4 Fraud Indicators & Risk Scoring Engine Module

Features:

- Rule-based anomaly flags, risk scores

Entities:

- **RiskIndicator**
 - IndicatorID
 - ClaimID
 - IndicatorType (HighCost/UnusualTiming/Pattern)
 - Severity
 - TriggeredDate
- **RiskScore**
 - ScoreID
 - ClaimID
 - ScoreValue
 - ComputedDate

4.5 Denial & Leakage Analysis Module

Features:

- Analyze denial codes, causes, error patterns

Entities:

- **DenialPattern**
 - PatternID
 - ClaimID
 - DenialCode
 - Reason
 - OccurrenceDate
- **LeakageFlag**
 - LeakageID
 - ClaimID
 - LeakageType (Overpayment/Delay/Error)
 - EstimatedLoss
 - IdentifiedDate

4.6 Adjuster & Operations Performance Analytics Module

Features:

- Productivity, workloads, SLA tracking

Entities:

- **AdjusterPerformance**
 - PerfID
 - AdjusterID
 - ClaimsHandled
 - AvgTAT
 - QualityScore
 - Period
- **SLAViolation**
 - ViolationID
 - ClaimID
 - ViolationType
 - ViolationDate

4.7 Cost, Reserve & Trend Analytics Module

Features:

- Trend analytics, tracking severity, reserves, aging

Entities:

- **ClaimCost**
 - CostID
 - ClaimID
 - CostType (Medical/Legal/Repair/Settlement)
 - Amount
 - CostDate
- **ClaimReserve**
 - ReserveID
 - ClaimID
 - ReserveAmount
 - UpdatedDate
- **AgingRecord**

- AgingID
- ClaimID
- AgingDays
- AgingBucket (0-30/31-60/61-90/90+)

4.8 Claims Dashboard, Reports & Visualization Module

Features:

- Reports for volumes, KPIs, risk, denials, reserves

Entities:

- **AnalyticsReport**
 - ReportID
 - Scope (Product/Region/ClaimType/Period)
 - Metrics
 - GeneratedDate

4.9 Notifications & Alerts Module

Features:

- Alerts for anomalies, denials spike, cost escalations

Entities:

- **Notification**
 - NotificationID
 - UserID
 - Message
 - Category (Risk/Denial/Cost/Performance)
 - Status (Unread/Read/Dismissed)
 - CreatedDate

5. Deployment Strategy

- **Local:** Angular/React UI + Spring Boot/.NET Core + relational DB
- **Production:** Containerized microservices, warehouse integration optional

6. Database Design

Tables:

- User(UserID, Name, Role, Email, Phone)
- AuditLog(AuditID, UserID, Action, Resource, Timestamp, Metadata)
- DataFeed(FeedID, FeedType, SourceSystem, LastSyncDate, Status)
- ClaimRaw(RawID, ClaimID, PayloadJSON, IngestedDate)
- ClaimKPI(KPIID, ClaimID, MetricName, MetricValue, MetricDate)
- RiskIndicator(IndicatorID, ClaimID, IndicatorType, Severity, TriggeredDate)
- RiskScore(ScoreID, ClaimID, ScoreValue, ComputedDate)
- DenialPattern(PatternID, ClaimID, DenialCode, Reason, OccurrenceDate)
- LeakageFlag(LeakageID, ClaimID, LeakageType, EstimatedLoss, IdentifiedDate)
- AdjusterPerformance(PerfID, AdjusterID, ClaimsHandled, AvgTAT, QualityScore, Period)
- SLAViolation(ViolationID, ClaimID, ViolationType, ViolationDate)
- ClaimCost(CostID, ClaimID, CostType, Amount, CostDate)
- ClaimReserve(ReserveID, ClaimID, ReserveAmount, UpdatedDate)
- AgingRecord(AgingID, ClaimID, AgingDays, AgingBucket)
- AnalyticsReport(ReportID, Scope, Metrics, GeneratedDate)
- Notification(NotificationID, UserID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **Claims Analytics Dashboard:** KPIs, TAT, reserves, volumes.
- **Fraud & Risk Dashboard:** Risk scores, red flags, hot claims.
- **Denial Dashboard:** Patterns, root causes, spikes.
- **Adjuster Productivity:** Workloads, quality scores, SLA violations.
- **Reserve & Cost Dashboard:** Severity, reserves, trends.
- **Admin Panel:** Data feeds, KPI definitions, user roles, reports.

8. Non-Functional Requirements

- **Performance:** Support large claim datasets; optimized analytical queries

- **Security:** RBAC, encryption, audit logging
- **Scalability:** Horizontal scaling for analytics workloads
- **Availability:** 99.9% uptime
- **Maintainability:** Modular microservices and versioned APIs
- **Observability:** Logs, metrics, anomaly indicators, performance dashboards

9. Assumptions & Constraints

- Phase-1 excludes live ML models (rule-based analytics only).
- Data ingestion uses batch/API integration; no ETL orchestration in Phase-1.
- Notifications are in-app only.
- Implementable using Java Spring Boot/ASP.NET Core, Angular/React with relational DB + optional data mart.